

Sean Liu

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EDUCATION

University of Washington

B.S. Informatics: Data Science, Software Engineering | Minor in Business

Seattle, WA

June 2027

TECHNICAL SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, HTML/CSS, SQL, R

Frameworks & Libraries: React.js, React Native, Next.js, Node.js, Express.js, Angular.js, PyTorch, TensorFlow, Pandas, Matplotlib, Altair, Mocha, Selenium

Developer Tools: PostgreSQL, GraphQL, MongoDB, Firebase, Git, Microsoft Azure, AWS, Figma, Agile Development

EXPERIENCE

Software Engineer Intern

Jun - Sept 2025

LightSys Technology Services

Colorado Springs, CO

- Developed a recurrent **LSTM autoencoder** in PyTorch to **automate anomaly detection** across **4.6M+** security logs, enabling customizable threshold tuning and log filtering through efficient batch processing
- Built a **preprocessing pipeline** to parse raw logs into sequential model input via **embeddings and vectorization**, while handling vocabularies of **2,000+ unique tokens** for certain log fields
- Contributed **~850 lines of modularized code** supporting end-to-end unsupervised training, inference, and evaluation, improving security and maintainability

PROJECTS

Savorly

Dec 2025

Python, PyTorch, PyTerrier, Streamlit

- Developed a hybrid recipe search and recommendation system over **3,000+ recipes** using **BM25 keyword search** and **PyTorch embeddings**
- Implemented an end-to-end retrieval pipeline with preprocessing, embedding generation, normalization, and hybrid ranking, deployed via **Streamlit**
- Added relevance tuning and personalization by aggregating a user's **last 10 searches** into a composite query for "for you" recommendations

HungryHusky

Dec 2025

Node.js, HTML/CSS, SQLite

- Built a full-stack restaurant reservation platform with a SQL-backed REST API (**20+ endpoints**), secure user authentication, **hashed password storage**, and persistent server-side sessions
- Implemented restaurant discovery and booking workflows across **30+ restaurants**, supporting multi-criteria search, availability validation, and **dynamic rating updates** from verified user reviews

Marathon Performance Analysis

Jun 2025

Python, Pandas, NumPy, Altair

- Cleaned and merged **250K+** marathon records across **350 years**, analyzing trends by year, event, age, and gender
- Created **interactive visualizations** through **Altair** displaying the impact environmental factors such as temperature and humidity on marathon performance

Data Analytics Dashboard

Mar 2025

TypeScript, Angular.js, GraphQL

- Implemented a **data analytics dashboard** to monitor student metrics for an online English tutoring platform with **150+ active users**
- Built real-time interactive charts in **TypeScript/Angular.js** to visualize student attendance and retention rates
- Utilized **GraphQL** and **Hasura** to query platform data and optimize database requests and retrieval speeds

Binccredible

Oct 2024

React Native, Flask, Figma

- Collaborated with a team of 4 to develop a mobile app for waste classification using **YOLO object detection** and **LLM-based description generation**.
- Designed and prototyped app interfaces in **Figma**, then implemented key **React Native** components to deliver a functional prototype during a 24-hour hackathon.